

Pricing American Put Options with Least-Squares Monte Carlo

Longstaff–Schwartz vs. Tsitsiklis–Van Roy under a shared benchmark design

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From European to American: what changes?

A European option can only be exercised at maturity

Its value depends on a single terminal payoff expectation

An American option can be exercised before maturity

Valuation therefore becomes both:

- a pricing problem
- a decision problem

American put pricing as an optimal stopping problem

We consider a vanilla American put with strike K and maturity T

Under the risk-neutral measure, the holder chooses an admissible stopping time τ

The time-0 value is

$$V_0 = \sup_{\tau \in \mathcal{T}} \mathbb{E}^{\mathbb{Q}} [e^{-r\tau} (K - S_{\tau})^+]$$

American option pricing is therefore an optimal stopping problem

Bellman recursion: exercise vs continuation

At each exercise date, the option value satisfies

$$V_i(S_{t_i}) = \max\{h(S_{t_i}), C_i(S_{t_i})\}$$

where

$$h(S_{t_i}) = (K - S_{t_i})^+$$

and

$$C_i(S_{t_i}) = \mathbb{E}^{\mathbb{Q}} \left[e^{-r\Delta t} V_{i+1}(S_{t_{i+1}}) \mid S_{t_i} \right]$$

- The holder compares immediate exercise with continuation at each date

What exactly is difficult to compute?

- The intrinsic value is easy to compute
- The main difficulty is the continuation value
- It is an unknown conditional expectation inside a backward recursion
- This is the computational issue that motivates the methods in the rest of the presentation

Why this project matters

American option pricing is an **optimal stopping** problem. Need both:

- a price,
- an exercise rule.

Direct trees / PDE methods work well in **simple 1D settings**.

Richer contracts quickly create problems with:

- larger state spaces,
- path dependence,
- changed dynamics.

This project studies two regression-based simulation methods under **one common design**.

Goal: compare methodology, not mismatched case studies.

Longstaff–Schwartz Paper Walkthrough

Simulates risk-neutral paths and works backward from maturity

Estimates continuation values by least-squares regression

Regression target: discounted **realized future cash flows**

Exercises when intrinsic value exceeds fitted continuation value

Usually regresses only on **in-the-money** paths

Tsitsiklis–Van Roy Paper Walkthrough

Treats the problem as approximate dynamic programming

Starts from the Bellman recursion and moves backward recursively

Approximates the value / continuation function with basis functions

Regression target: discounted **next-step estimated values**

Emphasizes approximation error and recursive stability

Methods compared (Josh)

Longstaff–Schwartz (LSM)

- Regress **discounted realized future cash flows**
- Builds a stopping rule from estimated continuation values
- Main practical pricing method in this project

Tsitsiklis–Van Roy (TVR)

- Regress **discounted next-step estimated values**
- Fitted value iteration / approximate dynamic programming
- Matched comparator under the same testbeds

$$V_i(s) = \max\{h(s), C_i(s)\}$$

Key difference: same Bellman problem, different recursion target.

Basis functions: how each method implements its regression (R.B)

Longstaff–Schwartz

Approximates the **continuation value**

$C_i(S_{t_i})$ using weighted Laguerre polynomials in normalised spot $X = S_t/K$:

$$L_0(X) = e^{-X/2}$$

$$L_1(X) = e^{-X/2}(1 - X)$$

$$L_2(X) = e^{-X/2}(1 - 2X + X^2/2)$$

- Applied to **in-the-money paths only**
- Regression target: **realized future cash flows**
- Basis is well-matched to stock price distribution on $[0, \infty)$

Tsitsiklis–Van Roy

Approximates the **value function**

$\tilde{Q}(x, r) = \sum_k r_k \phi_k(x)$ using polynomial basis. What matters is the best approximation error:

$$\epsilon_n^* = \min_r \|\Phi r - Q_n\|_{\pi_n}$$

- Applied to **all paths**
- Regression target: **discounted next-step value estimate**
- Theory is agnostic about basis family — quality determined by ϵ_n^*

Key difference: LSM uses realized data; TVR carries forward fitted estimates. Larger $\epsilon_n^* \Rightarrow$ more error accumulation backward through time.

Experimental Design: one benchmark layer, two extensions

The project is organized around three shared study layers motivated by corresponding examples in Longstaff–Schwartz paper.

Main validation layer: Vanilla American put under risk-neutral GBM

Shared inputs:

- $K = 40, r = 0.06$
- $S_0 \in \{36, 38, 40, 42, 44\}$
- $\sigma \in \{0.20, 0.40\}$
- $T \in \{1, 2\}$
- 50 exercise dates / year
- 100,000 paths (50,000 base + 50,000 antithetic)

Extensions:

- **Asian-style:** richer state vector (S_t, A_t)
- **Jump-style:** changed simulator dynamics

Benchmark hierarchy and implementation workflow

European BS lower reference → Direct American benchmarks (FD, CRR) →
Longstaff / TVR → Robustness + extensions

Black–Scholes European Benchmark (Lower Reference)

The Black–Scholes European put provides a strict theoretical floor for the American option, as a non-dividend-paying American put cannot be worth less than its European equivalent. For spot S_0 , strike K , volatility σ , risk-free rate r , and maturity T , the European price is:

$$P_{\text{Euro}} = Ke^{-rT}N(-d_2) - S_0N(-d_1),$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

This serves as a lower reference to quantify the early-exercise premium $(P_{\text{American}} - P_{\text{Euro}})$, rather than as the primary benchmark itself.

Implicit Finite-Difference Benchmark (Primary Reference)

The main direct benchmark is an implicit finite-difference solver, which directly evaluates the one-dimensional early-exercise problem on a discrete grid in stock price and time. The American put value $V(S, t)$ satisfies the Black–Scholes PDE in the continuation region:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0,$$

subject to the terminal payoff $V(S, T) = (K - S)^+$ and the early-exercise constraint $V(S, t) \geq (K - S)^+$. The spatial and time derivatives are discretized to produce a linear system. Following the backward step, the numerical value is projected onto the intrinsic value:

$$V_i^n = \max(V_i^{n,\text{cont}}, K - S_i),$$

where $V_i^{n,\text{cont}}$ is the continuation value at node (S_i, t_n) .

CRR Binomial Tree Benchmark (Auxiliary Check)

A Cox–Ross–Rubinstein (CRR) tree is included to ensure the internal coherence of the direct American benchmark layer. With time steps $\Delta t = T/N$, the stock moves are defined by $u = e^{\sigma\sqrt{\Delta t}}$ and $d = 1/u$, under the risk-neutral probability $p = \frac{e^{r\Delta t} - d}{u - d}$. The tree evaluates the terminal payoff $V_N(j) = (K - S_N(j))^+$ and is rolled backward via:

$$C_n(j) = e^{-r\Delta t}(pV_{n+1}(j+1) + (1-p)V_{n+1}(j)),$$

enforcing the early-exercise condition $V_n(j) = \max((K - S_n(j))^+, C_n(j))$ at each node.

Shared pieces wherever possible:

- simulator,
- basis construction,
- benchmark logic.

Different pieces:

- **LSM:** continuation-cash-flow regression,
- **TVR:** fitted value recursion.

In both methods, the fitted policy is then evaluated on fresh paths.

Why this matters: same environment, different recursion logic.

Vanilla benchmark: methods are sensible, but not interchangeable

Case	FD	LSMC (OOS)	TVR (OOS)	Pattern
$S_0 = 36, \sigma = 0.2, T = 1$	4.463	4.206	4.385	TVR closer
$S_0 = 40, \sigma = 0.2, T = 1$	2.299	2.444	2.278	LSM high, TVR close
$S_0 = 40, \sigma = 0.2, T = 2$	2.873	2.727	2.829	both reasonable
$S_0 = 40, \sigma = 0.4, T = 2$	6.897	6.971	6.859	both close
$S_0 = 44, \sigma = 0.4, T = 1$	3.924	4.132	3.838	LSM high, TVR low

Table 1: Partial Results of Vanilla American put pricing comparison under different spot, volatility, and maturity settings.

- Both methods produce **economically sensible American put values**.
- LSM can price **above or below** the FD benchmark.
- TVR tends to stay **near FD** and often slightly **below** it.

Takeaway: differences are methodological, not noise from different setups.

Vanilla visuals: benchmark alignment

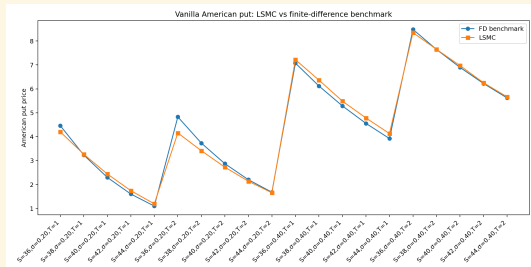


Figure 1: Longstaff benchmark vs finite difference.

LSM: tighter than same-path pricing, but still case-dependent around FD.

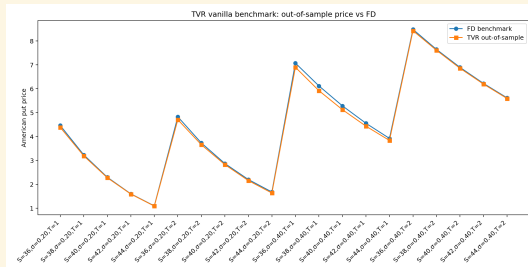


Figure 2: TVR benchmark vs finite difference.

TVR: clusters more closely around FD across the shared grid.

Validation matters: same-path optimism is mainly a Longstaff issue

Representative case	In-sample	Out-of-sample	Fit - Eval
LSM: $S_0 = 36$, $\sigma = 0.2$, $T = 1$	4.452	4.206	0.247
LSM: $S_0 = 40$, $\sigma = 0.2$, $T = 2$	3.060	2.739	0.321
LSM: $S_0 = 40$, $\sigma = 0.4$, $T = 2$	7.810	6.956	0.854
TVR: $S_0 = 36$, $\sigma = 0.2$, $T = 1$	4.386	4.381	0.005
TVR: $S_0 = 40$, $\sigma = 0.2$, $T = 2$	2.804	2.817	-0.014
TVR: $S_0 = 44$, $\sigma = 0.4$, $T = 1$	3.865	3.840	0.025

- LSM shows **clear same-path optimism**.
- TVR fit/eval gaps are **much smaller**.

Takeaway: out-of-sample policy evaluation is not optional.

Robustness snapshots

Test	Case	LSM pattern	TVR pattern
Paths(25000 $\rightarrow$ 200000)	(40, 0.2, 1)	2.317 $\rightarrow$ 2.447	2.270 $\rightarrow$ 2.282
Paths(25000 $\rightarrow$ 200000)	(40, 0.4, 2)	6.935 $\rightarrow$ 6.958	6.889 $\rightarrow$ 6.850
Grid(Ex. Dates/Yr 25 $\rightarrow$ 100)	(40, 0.2, 1)	2.561 $\rightarrow$ 2.344	2.295 $\rightarrow$ 2.281
Grid(Ex. Dates/Yr 25 $\rightarrow$ 100)	(40, 0.4, 2)	7.210 $\rightarrow$ 6.827	6.864 $\rightarrow$ 6.819
Basis(2 $\rightarrow$ 5)	(40, 0.2, 1)	2.385 $\rightarrow$ 2.458	2.151 $\rightarrow$ 2.285
Basis(2 $\rightarrow$ 5)	(40, 0.4, 2)	6.890 $\rightarrow$ 6.992	6.466 $\rightarrow$ 6.862

Table 2: Partial Results of In-sample versus Out-of-sample Policy Valuation for Representative Vanilla American Put Cases

- LSM is more sensitive to the interaction of fit and stopping-rule construction.
- TVR is more constrained by **under-parameterization** when the basis is too small.

Robustness: what moves the price?

Longstaff

- Path count reduces SE, but not monotone benchmark error
- Sensitive to exercise grid
- Sensitive to basis choice
- Seed variation is smaller than the main benchmark gaps

TVR

- Also benefits from more paths
- More stable across path budgets
- More directional improvement with richer bases
- Still mildly conservative against FD

Main message: price quality depends on more than path count:

- basis design,
- exercise grid,
- validation structure,
- approximation target.

Asian-style extension: American–Bermuda–Asian put

The state space is expanded to include both current spot and a running arithmetic average, with exercise restricted by a lockout period.

Under the risk-neutral measure, the stock follows GBM:

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}},$$

with constant r and σ . In the present Asian benchmark, the contract parameters are set to $K = 100$, $r = 0.06$, $\sigma = 0.20$, $T = 2$, with a lockout period of 0.25 years and 100 exercise dates per year in the discrete approximation. The same contract scope is used for both Longstaff and TVR.

To reflect the paper's idea that the average starts before valuation, we use a pre-started running arithmetic average:

$$A_t = \frac{\delta A_0 + \int_0^t S_u, du}{\delta + t}, \quad \delta = 0.25,$$

where A_0 is the initial average level specified in the benchmark grid. In the code, this is the second state variable carried along each simulated path.

The payoff from immediate exercise at time t is $h(S_t, A_t) = (K - A_t)^+$, so this is an average-based put rather than a spot-based vanilla put. Exercise is not allowed before the lockout date $t_{\text{lock}} = 0.25$. The time-zero value is therefore

$$V_0 = \sup_{\tau \in \mathcal{T}, \tau \geq t_{\text{lock}}} \mathbb{E}^{\mathbb{Q}} [e^{-r\tau} (K - A_\tau)^+].$$

With exercise dates $(0 = t_0 < t_1 < \dots < t_N = T)$, the discrete-time Bellman recursion is

$$V_i(s, a) = \max \left\{ (K - a)^+, \mathbb{E}^{\mathbb{Q}} \left[e^{-r\Delta t} V_{i+1}(S_{t_{i+1}}, A_{t_{i+1}}) \mid S_{t_i} = s, A_{t_i} = a \right] \right\}.$$

for dates after the lockout, with terminal condition $V_N(s, a) = (K - a)^+$.

Purpose: move from 1D vanilla validation to a richer continuation problem.

Asian-style results: both methods work, policies differ

Representative case	LSM (OOS)	LSM Fit-Eval	TVR (OOS)	TVR - LSM
$A_0 = 90, S_0 = 80$	13.777	1.016	17.498	3.720
$A_0 = 100, S_0 = 100$	3.213	0.122	3.598	0.384
$A_0 = 110, S_0 = 110$	1.066	0.068	1.078	0.013

Table 3: Partial results on the paper-inspired (A_0, S_0) grid with $K = 100$, $r = 0.06$, $\sigma = 0.2$, $T = 2$, $\text{lockout} = 0.25$, and 100 exercise dates per year.

- Both methods show the right comparative statics.
- LSM still has **positive fit-vs-eval gaps**.
- TVR is more **validation-stable** and often gives **higher values**.

Interpretation: multistate comparison only — not a direct PDE benchmark exercise.

Among all states the simulation visits at date t , where does the fitted policy choose to stop rather than continue?

Snapshot of $T = 2$, $A_0 = S_0 = 100$ at $t = 1$. x-axis = Spot S_t , y-axis = Average A_t , the color bar is intrinsic value at time t

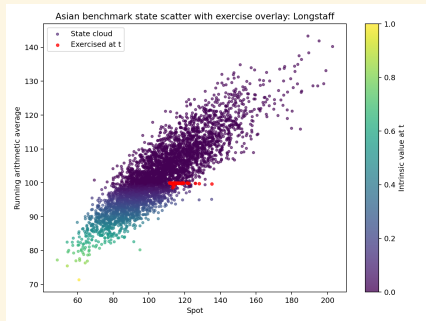


Figure 3: Longstaff Asian state cloud with exercise overlay.

More exercise-prone in borderline states

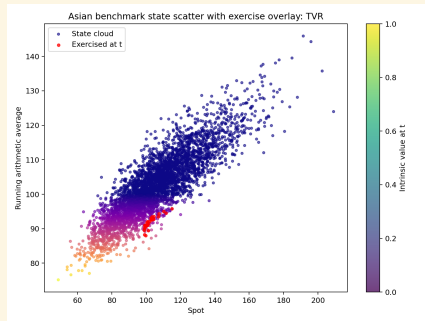


Figure 4: TVR Asian state cloud with exercise overlay.

More continuation-oriented

Jump-style extension: changed-dynamics study

The final layer replaces the standard GBM simulator with a stylized jump-to-ruin process.

Under the risk-neutral measure, we use a stylized jump-to-ruin process of the form

$$dS_t = (r + \lambda)S_t dt + \sigma S_t dW_t^{\mathbb{Q}} - S_{t-} dq_t,$$

where q_t is a Poisson process with intensity λ . When a jump occurs, the stock is sent to zero and remains there afterward. The $r + \lambda$ drift adjustment is the martingale correction used in the paper's jump comparison.

The immediate exercise payoff remains the vanilla American put payoff, $h(S_t) = (K - S_t)^+$, so the time-zero value is

$$V_0 = \sup_{\tau \in \mathcal{T}} \mathbb{E}^{\mathbb{Q}} [e^{-r\tau} (K - S_{\tau})^+].$$

With exercise dates $0 = t_0 < t_1 < \dots < t_N = T$, the discrete-time Bellman recursion is

$$V_i(s) = \max \left\{ (K - s)^+, \mathbb{E}^{\mathbb{Q}} [e^{-r\Delta t} V_{i+1}(S_{t_{i+1}}) \mid S_{t_i} = s] \right\}.$$

We compare two cases with the same strike and maturity, $K = 40$, $r = 0.06$, $T = 1$, using 26 exercise dates per year:

- **no jump:** $\lambda = 0, \sigma = 0.30$
- **jump-to-ruin:** $\lambda = 0.05, \sigma = 0.20$.

Purpose: test whether both methods remain coherent when dynamics change.

Jump-style results: changed dynamics, same methodological contrast

Case at $S_0 = 40$	American	Euro MC	Premium	Fit-Eval
LSM no-jump	4.093	3.570	0.523	0.235
LSM jump-to-ruin	3.614	3.220	0.394	0.149
TVR no-jump	3.743	3.588	0.155	0.001
TVR jump-to-ruin	3.277	3.217	0.060	-0.020

Table 4: Partial results on the matched jump grid with $K = 40$, $r = 0.06$, $T = 1$, 26 exercise dates per year, and 50,000 total paths. The no-jump case uses $\sigma = 0.30$, $\lambda = 0.00$, while the jump-to-ruin case uses $\sigma = 0.20$, $\lambda = 0.05$.

- Across the full jump grid, values fall with spot in both model cases.
- American values remain above European MC values.
- In this matched design, **jump-to-ruin values are generally lower** than no-jump values.
- TVR remains more conservative than LSM.

Jump-style visuals: price ranking + exercised-state summary

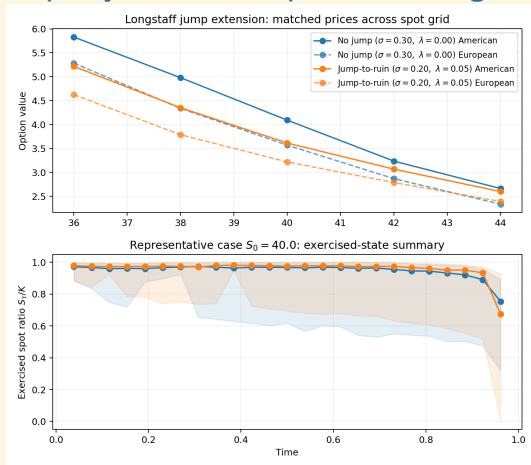


Figure 5: Longstaff matched jump extension

LSM: coherent under changed dynamics, but more exercise-prone.

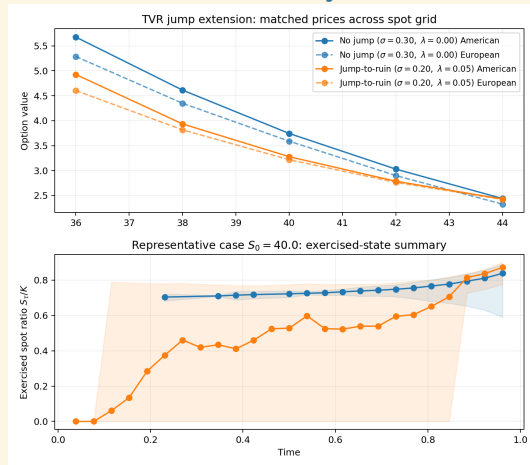


Figure 6: TVR matched jump extension

TVR: same ranking, more selective exercise; smaller fit/eval gap

Direct Longstaff vs TVR comparison (R.B)

Longstaff–Schwartz

- More directly tied to stopping-rule construction
- Strong practical simulation method
- More sensitive to:
 - fit/eval separation,
 - basis design,
 - exercise-grid design
- Can look optimistic if only same-path values are reported

Tsitsiklis–Van Roy

- More naturally read as fitted value iteration
- More stable under matched out-of-sample evaluation
- Often mildly conservative in benchmark alignment
- More continuation-oriented in the extensions

Bottom line: same optimal stopping problem, different numerical behavior.

Convergence theory: why the two methods behave differently (R.B)

LSM: Proposition 1 (downward bias)

$$V(X) \geq \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \text{LSM}(\omega_i; M, K)$$

LSM produces a stopping rule — asymptotically never overstates true value. But in finite samples with a finite basis, prices can sit above or below a numerical benchmark. The large fit/eval gaps in our results confirm that same-path evaluation overstates performance.

TVR: Theorem 1 (bounded error under trajectory sampling)

$$\tilde{e}_n^2 \leq \alpha^2 \tilde{e}_{n+1}^2 + (\epsilon_n^*)^2$$

If states are sampled from the risk-neutral trajectory distribution, error is bounded:

$$\tilde{e}_n \leq \frac{\epsilon}{\sqrt{2(1 - \alpha^2)}} \quad \text{independent of } N$$

Without trajectory sampling, errors grow **exponentially** with horizon.

TVR: Theorem 2 (almost sure convergence)

$$\hat{r}_n(m) \xrightarrow{a.s.} r_n \quad \text{as } m \rightarrow \infty$$

Sample-based regression converges to the idealised algorithm as paths grow.

This explains our results: TVR's small fit/eval gaps reflect bounded error accumulation. LSM's large gaps reflect in-sample optimism that out-of-sample evaluation corrects. Our basis sensitivity tables show ϵ_n^* shrinking as basis grows — exactly as Theorem 1 predicts.

Implementation Scope and Claims

The project does **not** claim strict numerical convergence proofs. Instead, it uses the papers' ideas to motivate a careful empirical workflow:

- direct benchmarking where possible,
- independent policy evaluation,
- matched testbeds across methods,
- robustness checks when approximation design matters.

Vanilla is the main validation layer.

Asian and jump are extension layers for:

- richer state representation,
- changed dynamics,
- workflow flexibility.

Testing claim: not production pricing — but defensible implementation evidence.

Limitations of the methods

GBM assumption:

- Constant volatility — real equity vol clusters and smiles
- No dividends — matters for equity options in practice
- Lognormal returns — fat tails and skew ignored
- Constant risk-free rate — unrealistic for long maturities

LSM method:

- Basis choice directly affects approximation quality
- In-sample optimism — same-path pricing overstates performance
- Finite paths leave residual Monte Carlo noise
- Exercise grid discretization introduces Bermudan approximation error

TVR comparator:

- Polynomial basis has larger ϵ_n^* than Laguerre basis
- All-path regression dilutes approximation in the ITM region
- Error accumulates backward through the recursion

Final takeaways (Josh)

Regression-based American pricing is **flexible and useful**.

But the output depends on more than simulation size:

- basis design,
- state representation,
- exercise grid,
- validation structure.

Longstaff and TVR are **not numerically interchangeable**.

Under the shared design here:

- **LSM** is more directly tied to practical stopping-rule construction,
- **TVR** is more validation-stable and mildly conservative.

Main conclusions

Regression-based methods are effective for pricing American-style options But pricing quality depends on more than simulation size Important design choices include:

- basis functions
- state representation
- exercise grid
- out-of-sample validation

Under a shared benchmark design, Longstaff and TVR are not numerically interchangeable

References

Longstaff, F. A., & Schwartz, E. S. (2001). *Valuing American options by simulation: A simple least-squares approach*.

Tsitsiklis, J. N., & Van Roy, B. (2001). *Regression methods for pricing complex American-style options*.